A counterexample to uniform L^2 stability in the Brascamp–Lieb variance inequality

Abstract

We show that the sharp uniform L^1 stability estimate for the Brascamp–Lieb variance inequality cannot be strengthened to an L^2 estimate with a constant depending only on the dimension. Already in dimension one we exhibit, for all sufficiently large S, a smooth strictly convex potential ϕ_S and a locally Lipschitz test function f_S such that

$$\frac{\mathrm{dist}_{L^2(\rho_{\phi_S})}(f_S, \mathcal{O}_{\mathrm{BL},\phi_S})^2}{\delta_{\mathrm{BL},\phi_S}(f_S)} = S\big(1 + O(S^{-1})\big) \to +\infty.$$

Equivalently, the constant in any putative inequality $\mathrm{dist}_{L^2} \le C\,\delta_{\mathrm{BL}}^{1/2}$ must blow up at rate at least $\sqrt{S}$.

1 Statement

Let $\phi : \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$ be an essentially continuous convex function with $0 < \int e^{-\phi} < \infty$. Write

$$Z_\phi := \int_{\mathbb{R}^d} e^{-\phi}\,dx, \qquad d\rho_\phi := Z_\phi^{-1} e^{-\phi}\,dx.$$

For smooth strictly convex ϕ, the *Brascamp–Lieb deficit* is

$$\delta_{\mathrm{BL},\phi}(f) = \int_{\mathbb{R}^d} \big\langle (D^2\phi)^{-1}\nabla f, \nabla f\big\rangle\,d\rho_\phi - \mathrm{Var}_{\rho_\phi}(f),$$

defined for any locally Lipschitz $f \in L^2(\rho_\phi)$. The Brascamp–Lieb inequality [2] asserts $\delta_{\mathrm{BL},\phi}(f) \ge 0$, with equality cases

$$\mathcal{O}_{\mathrm{BL},\phi} = \{a \cdot \nabla\phi + b : \ a \in \mathbb{R}^d,\ b \in \mathbb{R}\}.$$

When using $L^2(\rho_\phi)$ distances, this affine family is understood through its members that belong to $L^2(\rho_\phi)$; for the explicit potentials constructed below all the listed generators are in L^2.

Theorem 1.1. *There is no finite constant C such that*

$$\mathrm{dist}_{L^2(\rho_\phi)}(f, \mathcal{O}_{\mathrm{BL},\phi}) \le C\,\delta_{\mathrm{BL},\phi}(f)^{1/2} \tag{1}$$

holds simultaneously for every smooth strictly convex $\phi : \mathbb{R} \to \mathbb{R}$ with $\int e^{-\phi} < \infty$ and every locally Lipschitz $f \in L^2(\rho_\phi)$. Consequently, no dimension-only constant $C = C(d)$ can exist in any dimension $d \geq 1$.

Spectral interpretation

In dimension one, the Brascamp–Lieb Dirichlet form is

$$\mathcal{E}_\phi(f,g) := \int_{\mathbb{R}} \frac{f'g'}{\phi''}\, d\rho_\phi, \qquad \mathcal{E}_\phi(f) := \mathcal{E}_\phi(f,f). \tag{2}$$

Here the form domain means the locally absolutely continuous functions $f \in L^2(\rho_\phi)$ for which $\mathcal{E}_\phi(f) < \infty$. After closing the form, its restriction to the orthogonal complement of constants in $L^2(\rho_\phi)$ defines a non-negative self-adjoint operator L_ϕ via $\mathcal{E}_\phi(f,g) = \langle L_\phi f, g\rangle_{L^2(\rho_\phi)}$, and the Brascamp–Lieb inequality is the assertion that $\inf \operatorname{spec}\big(L_\phi \restriction \{1\}^\perp\big) \geq 1$. Theorem 1.1 is the assertion that the spectral gap $\inf \operatorname{spec}\big(L_\phi \restriction \mathcal{O}^\perp_{\mathrm{BL},\phi}\big) - 1$ *cannot* be bounded below by a positive universal constant: the construction below produces potentials for which the Rayleigh quotient on $\mathcal{O}^\perp_{\mathrm{BL},\phi}$ approaches 1 from above. If one also knows compactness of the resolvent for the corresponding one-dimensional operator, this can be phrased as collapse of the next eigenvalue.

The basic optimizer direction at the lowest level is ϕ' itself.

Lemma 1.2. *Let $\phi : \mathbb{R} \to \mathbb{R}$ be smooth and strictly convex, with $0 < \int e^{-\phi} < \infty$. Assume in addition that ϕ'' is bounded and $\phi' \in L^2(\rho_\phi)$; in particular, this holds for the potentials ϕ_S constructed below. Then $\int \phi'\, d\rho_\phi = 0$, ϕ' belongs to the form domain, and*

$$\mathcal{E}_\phi(\phi', g) = \langle \phi', g\rangle_{L^2(\rho_\phi)} \qquad \textit{for every locally Lipschitz form-domain } g.$$

In particular $\mathcal{E}_\phi(\phi') = \mathrm{Var}_{\rho_\phi}(\phi')$, and ϕ' satisfies the weak eigenvalue identity at level 1 against locally Lipschitz form-domain test functions.

Proof. Integrability of $e^{-\phi}$ on $\mathbb{R}$ together with strict convexity forces $\phi(x) \to +\infty$ as $|x| \to \infty$, hence $e^{-\phi(\pm R)} \to 0$ as $R \to \infty$. Moreover, by convexity ϕ' is monotone non-decreasing, so $\phi'(x)$ has a (possibly infinite) limit as $x \to \pm\infty$, and there exists $x_0 > 0$ with $\phi'(x) \geq 0$ for $x \geq x_0$ and $\phi'(x) \leq 0$ for $x \leq -x_0$ (otherwise ϕ could not satisfy $\phi(\pm\infty) = +\infty$). Thus

$$\int_{\mathbb{R}} |\phi'(x)| e^{-\phi(x)}\, dx = -\int_{-\infty}^{-x_0} (e^{-\phi})'\, dx + \int_{-x_0}^{x_0} |\phi'| e^{-\phi}\, dx + \int_{x_0}^{\infty} -(e^{-\phi})'\, dx < \infty,$$

and integrating $\phi' e^{-\phi} = -(e^{-\phi})'$ over the two tails and then letting the endpoints tend to infinity gives $\int \phi'\, d\rho_\phi = 0$.

Integration by parts for general h. Let $h \in L^2(\rho_\phi)$ be locally Lipschitz and in the form domain. Since ϕ'' is bounded, $h' \in L^2(\rho_\phi)$. By Cauchy–Schwarz, $h\phi' e^{-\phi}, h e^{-\phi}, h' e^{-\phi} \in L^1(\mathbb{R})$. Choose sequences $R_k \to +\infty$ and $R'_k \to -\infty$

along which $|h(R_k)|e^{-\phi(R_k)} + |h(R'_k)|e^{-\phi(R'_k)} \to 0$. Integration by parts on $[R'_k, R_k]$ gives

$$\int_{R'_k}^{R_k} h\phi' e^{-\phi}\,dx = -\int_{R'_k}^{R_k} h(e^{-\phi})'\,dx = -\big[h\,e^{-\phi}\big]_{R'_k}^{R_k} + \int_{R'_k}^{R_k} h' e^{-\phi}\,dx.$$

The boundary term vanishes along the chosen sequence; passing to the limit yields

$$\int_{\mathbb{R}} h\phi' e^{-\phi}\,dx = \int_{\mathbb{R}} h' e^{-\phi}\,dx. \tag{3}$$

For any locally Lipschitz form-domain g, applying (3) to $h = g$ yields

$$\mathcal{E}_\phi(\phi', g) = \int \frac{(\phi')' g'}{\phi''}\,d\rho_\phi = \int g'\,d\rho_\phi = \int g\,\phi'\,d\rho_\phi = \langle \phi', g\rangle_{L^2(\rho_\phi)}.$$

Specializing to $g = \phi'$ gives $\mathcal{E}_\phi(\phi') = \|\phi'\|^2_{L^2(\rho_\phi)} = \mathrm{Var}_{\rho_\phi}(\phi')$, since ϕ' has ρ_ϕ-mean zero.

For the later potentials ϕ_S, the assumptions are immediate from the explicit formulas: ϕ''_S is bounded for each fixed S, while on the tails $\phi'_S(x) = \eta x \pm S$ and $e^{-\phi_S(x)}$ has at least Gaussian decay. □

2 The construction

Fix a non-negative even C^∞ bump $\kappa : \mathbb{R} \to [0,\infty)$ supported in $[-1,1]$ and normalized by $\int_{\mathbb{R}} \kappa(u)\,du = 1$. For parameters $S \geq 1$, $\varepsilon \in (0, \frac{1}{4})$ and $\eta > 0$ define

$$\phi''_{S,\varepsilon,\eta}(x) := \eta + \frac{S}{\varepsilon}\kappa\left(\frac{x-1}{\varepsilon}\right) + \frac{S}{\varepsilon}\kappa\left(\frac{x+1}{\varepsilon}\right), \tag{4}$$

together with the normalization $\phi_{S,\varepsilon,\eta}(0) = \phi'_{S,\varepsilon,\eta}(0) = 0$. The right-hand side of (4) is C^∞, even, and bounded below by $\eta > 0$, so $\phi_{S,\varepsilon,\eta}$ is C^∞, even, strictly convex, with $\phi_{S,\varepsilon,\eta}(x) \geq \eta x^2/2$, hence $\int e^{-\phi_{S,\varepsilon,\eta}} < \infty$. By construction, $\phi'_{S,\varepsilon,\eta}$ is odd.

We require the parameters to satisfy

$$S\varepsilon \to 0, \qquad \eta\, S^{-2} \to 0, \qquad \eta \to 0, \qquad \text{as } S \to \infty. \tag{5}$$

For concreteness we work throughout with the explicit choice

$$\varepsilon = S^{-3}, \qquad \eta = S^{-4}, \tag{6}$$

which obeys all conditions in (5), and abbreviate

$$\phi_S := \phi_{S,S^{-3},S^{-4}}, \qquad \rho_S := \rho_{\phi_S}, \qquad Z_S := \int_{\mathbb{R}} e^{-\phi_S}\,dx.$$

All asymptotic notation $O(\,\cdot\,)$, $o(1)$ below is understood as $S \to \infty$.

Notation for the three regions

Decompose $\mathbb{R}$ into the central plateau, the two transition layers, and the exterior:

$$C_S := [-1+\varepsilon, 1-\varepsilon], \quad T_S := I_S^+ \cup I_S^-, \quad E_S := \mathbb{R} \setminus (C_S \cup T_S),$$

where

$$I_S^+ := (1-\varepsilon, 1+\varepsilon), \qquad I_S^- := (-1-\varepsilon, -1+\varepsilon).$$

Profile of ϕ_S, ϕ_S' and the partition function

Direct integration of (4), together with $\int \kappa = 1$ and the support assumption on κ, gives:

- On C_S: $\phi_S'' \equiv \eta$, so $\phi_S'(x) = \eta x$ and $\phi_S(x) = \eta x^2/2 \in [0, \eta/2]$.
- On I_S^+: ϕ_S' is non-decreasing and $\int_{1-\varepsilon}^{1+\varepsilon} \phi_S''\, dx = 2\eta\varepsilon + S$, so ϕ_S' ranges from $\eta(1-\varepsilon)$ at the left endpoint to $\eta(1+\varepsilon) + S$ at the right endpoint.
- On $(1+\varepsilon, \infty)$: $\phi_S'' = \eta$, so $\phi_S'(x) = \eta x + S$.
- Symmetric statements hold on I_S^- and $(-\infty, -1-\varepsilon)$.

Lemma 2.1. *Under the choice* (6)*, for all sufficiently large S:*

(a) $0 \le \phi_S(x) \le \eta/2 = \frac{1}{2}S^{-4}$ *for* $x \in C_S$.

(b) $0 \le \phi_S(x) \le \frac{\eta}{2} + 2\varepsilon\big(\eta + S\big) = O(S\varepsilon) = O(S^{-2})$ *for* $x \in T_S$*. In particular* $e^{\phi_S(x)} = 1 + O(S^{-2})$ *uniformly on* T_S.

(c) For $x > 1+\varepsilon$,

$$\phi_S(x) = \phi_S(1+\varepsilon) + S\,(x - 1 - \varepsilon) + \tfrac{\eta}{2}\big(x^2 - (1+\varepsilon)^2\big).$$

(d) $\displaystyle\int_{I_S^+} \phi_S''(t)\, e^{\phi_S(t)}\, dt = S\big(1 + O(S^{-2})\big).$

Proof. (a) is immediate from $\phi_S(x) = \eta x^2/2$ and $|x| \le 1$.

(b) For $x \in I_S^+$,

$$\phi_S(x) = \phi_S(1-\varepsilon) + \int_{1-\varepsilon}^{x} \phi_S'(t)\, dt \le \tfrac{\eta}{2} + 2\varepsilon \sup_{I_S^+} \phi_S' \le \tfrac{\eta}{2} + 2\varepsilon(\eta + S).$$

Substituting $\eta = S^{-4}$, $\varepsilon = S^{-3}$ gives the bound $O(S\varepsilon) = O(S^{-2})$. The bound $|e^{\phi_S} - 1| \le \phi_S e^{\phi_S} = O(S^{-2})$ follows. The bound on I_S^- is identical by even symmetry.

(c) For $x > 1+\varepsilon$, $\phi_S(x) - \phi_S(1+\varepsilon) = \int_{1+\varepsilon}^{x} \phi_S'(t)\, dt = \int_{1+\varepsilon}^{x} (\eta t + S)\, dt$, which evaluates to the stated expression.

(d) Using $e^{\phi_S} = 1 + O(S^{-2})$ uniformly on I_S^+ from (b) and $\int_{I_S^+} \phi_S'' \, dx = S + 2\eta\varepsilon = S + O(S^{-7})$, we get

$$\int_{I_S^+} \phi_S'' e^{\phi_S} \, dx = \int_{I_S^+} \phi_S'' \, dx + \int_{I_S^+} \phi_S'' \big(e^{\phi_S} - 1\big) \, dx = S + O(S^{-7}) + O(S^{-2}) \cdot S = S\big(1 + O(S^{-2})\big).$$

□

Lemma 2.2. *Under the choice* (6)*, as* $S \to \infty$*:*

(a) $\displaystyle\int_{1+\varepsilon}^{\infty} e^{-\phi_S(x)} \, dx = \frac{1}{S} + O(S^{-3})$.

(b) $Z_S = 2 + \dfrac{2}{S} + O(S^{-3})$.

(c) Set $q_S := \rho_S(E_S) = \rho_S\big(\{|x| > 1 + \varepsilon\}\big)$ *and* $t_S := \rho_S(T_S)$*. Then*

$$q_S = \frac{1}{S} - \frac{1}{S^2} + O(S^{-3}), \qquad t_S = O(S^{-3}).$$

Proof. (a) Substitute $u = x - 1 - \varepsilon$ in part (c) of Lemma 2.1, set $\widetilde{S} := S + \eta(1 + \varepsilon) = S\big(1 + O(\eta/S)\big)$, and use $\phi_S(1 + \varepsilon) = O(S^{-2})$:

$$\int_{1+\varepsilon}^{\infty} e^{-\phi_S} \, dx = e^{-\phi_S(1+\varepsilon)} \int_0^{\infty} e^{-\widetilde{S}\, u - \eta u^2/2} \, du.$$

The elementary bound $0 \le 1 - e^{-v} \le v$ for $v \ge 0$ gives

$$0 \le \frac{1}{\widetilde{S}} - \int_0^{\infty} e^{-\widetilde{S}\, u - \eta u^2/2} \, du \le \frac{\eta}{2} \int_0^{\infty} u^2 e^{-\widetilde{S} u} \, du = \frac{\eta}{\widetilde{S}^3}. \tag{7}$$

Since $\widetilde{S} = S + \eta(1 + \varepsilon)$, this implies

$$\int_0^{\infty} e^{-\widetilde{S}\, u - \eta u^2/2} \, du = \frac{1}{S} + O\Big(\frac{\eta}{S^2} + \frac{\eta}{S^3}\Big) = \frac{1}{S} + O(S^{-6}).$$

Combined with $e^{-\phi_S(1+\varepsilon)} = 1 + O(S^{-2})$ this gives

$$\int_{1+\varepsilon}^{\infty} e^{-\phi_S} \, dx = \big(1 + O(S^{-2})\big)\big(1/S + O(S^{-6})\big) = \frac{1}{S} + O(S^{-3}),$$

proving (a).

(b) Decompose $\int_0^{\infty} e^{-\phi_S} \, dx$ over the three regions $[0, 1 - \varepsilon]$, $[1 - \varepsilon, 1 + \varepsilon]$, $[1 + \varepsilon, \infty)$.

On $[0, 1 - \varepsilon]$: $\phi_S(x) = \eta x^2/2$, so $\int_0^{1-\varepsilon} e^{-\eta x^2/2} \, dx = (1 - \varepsilon) + O(\eta) = 1 - \varepsilon + O(S^{-4})$.

On $[1 - \varepsilon, 1 + \varepsilon]$: $e^{-\phi_S} = 1 + O(S^{-2})$ by Lemma 2.1(b), so $\int = 2\varepsilon + O(\varepsilon/S^2) = 2\varepsilon + O(S^{-5})$.

On $[1 + \varepsilon, \infty)$: by part (a), $\int = 1/S + O(S^{-3})$.

Adding and using $\varepsilon = S^{-3}$:

$$\int_0^\infty e^{-\phi_S}\,dx = 1 - \varepsilon + 2\varepsilon + \frac{1}{S} + O(S^{-3}) = 1 + \frac{1}{S} + O(S^{-3}).$$

Multiplying by 2 (even symmetry) yields $Z_S = 2 + 2/S + O(S^{-3})$.

(c) The layer mass is bounded by the Lebesgue measure of T_S times $\sup_{T_S} e^{-\phi_S}/Z_S \le 4\varepsilon/Z_S = O(\varepsilon) = O(S^{-3})$. For q_S:

$$q_S = \frac{2}{Z_S}\int_{1+\varepsilon}^\infty e^{-\phi_S}\,dx = \frac{2/S + O(S^{-3})}{2 + 2/S + O(S^{-3})} = \frac{1}{S}\cdot\frac{1 + O(S^{-2})}{1 + 1/S + O(S^{-3})}.$$

Expanding the geometric series for $1/(1 + 1/S + O(S^{-3})) = 1 - 1/S + 1/S^2 + O(S^{-3})$:

$$q_S = \frac{1}{S}\Big(1 - \frac{1}{S} + \frac{1}{S^2} + O(S^{-3})\Big)\big(1 + O(S^{-2})\big) = \frac{1}{S} - \frac{1}{S^2} + O(S^{-3}). \qquad \square$$

The test function g_S

Define $g_S : \mathbb{R} \to [0,1]$ by

$$g_S(x) := \begin{cases} 0 & x \in C_S, \\ \dfrac{\int_{1-\varepsilon}^{|x|} \phi_S''(t) e^{\phi_S(t)}\,dt}{\int_{I_S^+} \phi_S''(t) e^{\phi_S(t)}\,dt} & |x| \in (1-\varepsilon, 1+\varepsilon), \\ 1 & x \in E_S. \end{cases} \tag{8}$$

On each of C_S and E_S, g_S is constant; on each layer it is (a reparametrization of) the energy-minimizing interpolation between the prescribed boundary values, as we verify in Lemma 2.3 below.

Continuity at the four endpoints $\pm 1 \pm \varepsilon$ is built in: at $|x| = 1 - \varepsilon$ the layer formula gives $g_S = 0$, matching C_S; at $|x| = 1 + \varepsilon$ it gives $g_S = 1$, matching E_S. By construction g_S is even, since the formula depends only on $|x|$.

Since ϕ_S'' is bounded on $\mathbb{R}$ (it has supremum $\eta + (S/\varepsilon)\|\kappa\|_\infty = O(S^4)$), g_S' is bounded on each layer, hence g_S is globally Lipschitz. In particular, $g_S \in L^\infty(\mathbb{R}) \subset L^2(\rho_S)$ and is locally Lipschitz.

Lemma 2.3. *The Brascamp–Lieb energy of g_S is*

$$\mathcal{E}_{\phi_S}(g_S) = \frac{2}{Z_S \int_{I_S^+} \phi_S''(t) e^{\phi_S(t)}\,dt} = \frac{2}{Z_S\, S}\big(1 + O(S^{-2})\big).$$

Proof. We first compute the energy of the optimal interpolant on a single layer. Let $I = (a,b)$ be an interval and consider the variational problem

$$J^* := \inf\left\{\int_I \frac{(h'(x))^2}{\phi_S''(x)}\, e^{-\phi_S(x)}\,dx : \ h \in W^{1,2}(I),\ h(a) = 0,\ h(b) = \Delta\right\}, \tag{9}$$

where the boundary values are interpreted as traces. Setting $w(x) := e^{-\phi_S(x)}/\phi_S''(x) > 0$, the integrand is $(h')^2 w$. The Euler–Lagrange equation $(h'\,w)' = 0$ gives $h'(x) = c/w(x) = c\,\phi_S''(x)\,e^{\phi_S(x)}$ for some constant c. The boundary condition $\int_a^b h'\,dx = \Delta$ forces

$$c = \frac{\Delta}{\int_I \phi_S''\,e^{\phi_S}\,dx}.$$

Substituting $h' = c\phi_S'' e^{\phi_S}$ into the integral gives

$$\int_I (h')^2 w\,dx = c^2 \int_I \frac{(\phi_S'')^2 e^{2\phi_S}}{\phi_S''}\,e^{-\phi_S}\,dx = c^2 \int_I \phi_S'' e^{\phi_S}\,dx = \frac{\Delta^2}{\int_I \phi_S'' e^{\phi_S}\,dx}.$$

This is in fact the minimum: the integrand is a coercive non-negative quadratic form in h', with a strictly positive weight w bounded above and below on the compact interval $\overline{I} = [a,b]$ (here w is bounded above by $1/\eta$ and below by $\varepsilon/(S\|\kappa\|_\infty\,e^{O(S^{-2})})$, both strictly positive). The set of admissible h is the affine subspace $\{h_0 + W_0^{1,2}(I)\}$ with h_0 a fixed admissible interpolant, and J is strictly convex coercive on this subspace, so a minimizer exists and is unique. The Euler–Lagrange formula above gives that minimizer.

By construction (8), g_S restricted to I_S^+ has $g_S(1-\varepsilon) = 0$, $g_S(1+\varepsilon) = 1$, and $g_S'(x) = \phi_S''(x)e^{\phi_S(x)} \big/ \int_{I_S^+} \phi_S'' e^{\phi_S}$, which is exactly the Euler–Lagrange minimizer with $\Delta = 1$. The same holds on I_S^- with $\Delta = -1$ (since g_S ranges from 1 at $-1-\varepsilon$ to 0 at $-1+\varepsilon$, sign of Δ is opposite, but $\Delta^2 = 1$ in either case). Outside the layers g_S is constant, so $g_S' \equiv 0$ on $C_S \cup E_S$. Therefore

$$\mathcal{E}_{\phi_S}(g_S) = \frac{1}{Z_S}\left(\int_{I_S^+} \frac{(g_S')^2}{\phi_S''} e^{-\phi_S}\,dx + \int_{I_S^-} \frac{(g_S')^2}{\phi_S''} e^{-\phi_S}\,dx\right) = \frac{1}{Z_S}\big(J^*_{I_S^+} + J^*_{I_S^-}\big),$$

and by the formula above with $\Delta^2 = 1$ on each layer and the symmetry $\int_{I_S^+} \phi_S'' e^{\phi_S} = \int_{I_S^-} \phi_S'' e^{\phi_S}$,

$$\mathcal{E}_{\phi_S}(g_S) = \frac{2}{Z_S \int_{I_S^+} \phi_S''\,e^{\phi_S}\,dx}.$$

The asymptotic statement now follows from Lemma 2.1(d). □

Lemma 2.4. *The variance of g_S satisfies*

$$\mathrm{Var}_{\rho_S}(g_S) = q_S(1-q_S) + O(S^{-3}) = \frac{1}{S} - \frac{2}{S^2} + O(S^{-3}).$$

Proof. On E_S, $g_S \equiv 1$; on C_S, $g_S \equiv 0$; on T_S, $0 \le g_S \le 1$. Hence

$$\int g_S\,d\rho_S = \rho_S(E_S) + \int_{T_S} g_S\,d\rho_S = q_S + \mathcal{R}_1, \qquad |\mathcal{R}_1| \le t_S = O(S^{-3}),$$

$$\int g_S^2\,d\rho_S = \rho_S(E_S) + \int_{T_S} g_S^2\,d\rho_S = q_S + \mathcal{R}_2, \qquad |\mathcal{R}_2| \le t_S = O(S^{-3}),$$

by Lemma 2.2(c). Therefore

$$\mathrm{Var}_{\rho_S}(g_S) = (q_S+\mathcal{R}_2)-(q_S+\mathcal{R}_1)^2 = q_S-q_S^2+\big(\mathcal{R}_2-2q_S\mathcal{R}_1-\mathcal{R}_1^2\big) = q_S(1-q_S)+O(S^{-3}),$$

where in the last step we used $q_S = O(S^{-1})$ to absorb $-2q_S\mathcal{R}_1 = O(S^{-4})$ and $\mathcal{R}_1^2 = O(S^{-6})$ into the $O(S^{-3})$ error.

Substituting $q_S = 1/S - 1/S^2 + O(S^{-3})$ from Lemma 2.2(c),

$$q_S^2 = \frac{1}{S^2}\Big(1 - \frac{1}{S} + O(S^{-2})\Big)^2 = \frac{1}{S^2} - \frac{2}{S^3} + O(S^{-4}),$$

hence

$$q_S(1-q_S) = q_S - q_S^2 = \Big(\tfrac{1}{S} - \tfrac{1}{S^2}\Big) - \Big(\tfrac{1}{S^2} - \tfrac{2}{S^3}\Big) + O(S^{-3}) = \tfrac{1}{S} - \tfrac{2}{S^2} + O(S^{-3}). \quad \square$$

3 The spectral gap above $\mathcal{O}_{\mathrm{BL},\phi_S}$ closes

Let

$$c_S := \int_{\mathbb{R}} g_S \, d\rho_S, \qquad f_S := g_S - c_S.$$

Since g_S is even and ρ_S is even, $c_S \in \mathbb{R}$ and f_S is even with ρ_S-mean zero. The Brascamp–Lieb deficit and the variance are unchanged by additive constants, so

$$\mathcal{E}_{\phi_S}(f_S) = \mathcal{E}_{\phi_S}(g_S), \qquad \mathrm{Var}_{\rho_S}(f_S) = \mathrm{Var}_{\rho_S}(g_S), \qquad \delta_{\mathrm{BL},\phi_S}(f_S) = \delta_{\mathrm{BL},\phi_S}(g_S). \tag{10}$$

Lemma 3.1. *f_S is orthogonal in $L^2(\rho_S)$ to the optimizer space $\mathcal{O}_{\mathrm{BL},\phi_S} = \mathrm{span}_{\mathbb{R}}\{1, \phi_S'\}$. Consequently*

$$\mathrm{dist}_{L^2(\rho_S)}(f_S, \mathcal{O}_{\mathrm{BL},\phi_S})^2 = \|f_S\|^2_{L^2(\rho_S)} = \mathrm{Var}_{\rho_S}(g_S).$$

Proof. $\langle f_S, 1\rangle_{L^2(\rho_S)} = \int f_S \, d\rho_S = 0$ by construction. For the second generator, since f_S is even, ϕ_S' is odd, and $e^{-\phi_S}$ is even, the integrand $f_S(x)\phi_S'(x)e^{-\phi_S(x)}$ is an odd function in $L^1(\mathbb{R})$, so

$$\langle f_S, \phi_S'\rangle_{L^2(\rho_S)} = \frac{1}{Z_S}\int_{\mathbb{R}} f_S(x)\phi_S'(x)e^{-\phi_S(x)}\, dx = 0.$$

Since ϕ_S' is non-constant (it ranges over an interval containing $\pm S$), the two generators 1 and ϕ_S' are linearly independent in $L^2(\rho_S)$, so $f_S \perp \mathcal{O}_{\mathrm{BL},\phi_S}$. The distance formula follows from $\|f_S\|^2 = \int f_S^2 \, d\rho_S = \mathrm{Var}_{\rho_S}(g_S)$, since f_S has ρ_S-mean zero. $\square$

Proposition 3.2. *With the choice* (6)*, as $S \to \infty$,*

$$\mathcal{E}_{\phi_S}(f_S) = \frac{1}{S} - \frac{1}{S^2} + O(S^{-3}), \qquad \mathrm{Var}_{\rho_S}(f_S) = \frac{1}{S} - \frac{2}{S^2} + O(S^{-3}),$$

and consequently

$$\delta_{\mathrm{BL},\phi_S}(f_S) = \frac{1}{S^2} + O(S^{-3}).$$

In particular $\delta_{\mathrm{BL},\phi_S}(f_S) > 0$ *for all sufficiently large* S*, and the Brascamp–Lieb Rayleigh quotient* $\mathcal{E}_{\phi_S}(f_S)/\operatorname{Var}_{\rho_S}(f_S)$ *converges to* 1 *from above at rate* $1/S$*.*

Proof. Combining Lemma 2.3 with the identity $2/(Z_S\,S) = q_S \cdot \big(1 + O(S^{-2})\big)$ from Lemma 2.2(c) (alternatively, computing directly: $2/(Z_S S) = 1/(S + 1 + O(S^{-2})) = 1/S - 1/S^2 + O(S^{-3})$),

$$\mathcal{E}_{\phi_S}(g_S) = \frac{2}{Z_S\,S}\big(1{+}O(S^{-2})\big) = \Big(\tfrac{1}{S}{-}\tfrac{1}{S^2}{+}O(S^{-3})\Big)\big(1{+}O(S^{-2})\big) = \tfrac{1}{S}{-}\tfrac{1}{S^2}{+}O(S^{-3}).$$

Lemma 2.4 gives the variance asymptotics. Subtracting,

$$\delta_{\mathrm{BL},\phi_S}(f_S) = \mathcal{E}_{\phi_S}(g_S){-}\operatorname{Var}_{\rho_S}(g_S) = \Big(\tfrac{1}{S}{-}\tfrac{1}{S^2}\Big){-}\Big(\tfrac{1}{S}{-}\tfrac{2}{S^2}\Big){+}O(S^{-3}) = \tfrac{1}{S^2}{+}O(S^{-3}).$$

The displayed asymptotic implies $\delta_{\mathrm{BL},\phi_S}(f_S) > 0$ for all sufficiently large S. □

Theorem 3.3 (1D part of Theorem 1.1)**.** *With the choice* (6) *and the test function* f_S *defined above,*

$$\frac{\operatorname{dist}_{L^2(\rho_S)}(f_S, \mathcal{O}_{\mathrm{BL},\phi_S})^2}{\delta_{\mathrm{BL},\phi_S}(f_S)} = S\big(1 + O(S^{-1})\big) \xrightarrow[S\to\infty]{} +\infty.$$

Therefore the constant in any putative inequality (1) *must blow up at rate at least* $\sqrt{S}$*.*

Proof. By Lemma 3.1 and Lemma 2.4,

$$\operatorname{dist}_{L^2(\rho_S)}(f_S, \mathcal{O}_{\mathrm{BL},\phi_S})^2 = \operatorname{Var}_{\rho_S}(g_S) = \tfrac{1}{S} - \tfrac{2}{S^2} + O(S^{-3}).$$

Combined with $\delta_{\mathrm{BL},\phi_S}(f_S) = 1/S^2 + O(S^{-3})$ from Proposition 3.2,

$$\frac{\operatorname{dist}^2}{\delta_{\mathrm{BL}}} = \frac{1/S - 2/S^2 + O(S^{-3})}{1/S^2 + O(S^{-3})} = \frac{(1/S)\big(1 - 2/S + O(S^{-2})\big)}{(1/S^2)\big(1 + O(S^{-1})\big)} = S\big(1 + O(S^{-1})\big).$$

Equivalently $\operatorname{dist}/\delta_{\mathrm{BL}}^{1/2} = \sqrt{S}\,\big(1 + O(S^{-1})\big)$, so any constant in (1) must satisfy $C \geq \sqrt{S}\,(1 + o(1))$. □

Higher dimensions

Fix $d \geq 2$ and define the product potential

$$\Phi_S(x, y) := \phi_S(x) + \frac{|y|^2}{2}, \qquad (x, y) \in \mathbb{R} \times \mathbb{R}^{d-1}.$$

Then Φ_S is C^∞, strictly convex (as a sum of strictly convex functions in disjoint variables), with $D^2\Phi_S = \operatorname{diag}(\phi_S''(x), I_{d-1})$ and $(D^2\Phi_S)^{-1} = \operatorname{diag}(1/\phi_S''(x), I_{d-1})$.

The associated Gibbs measure is the product $\rho_{\Phi_S} = \rho_S \otimes \gamma_{d-1}$, with γ_{d-1} the standard Gaussian on $\mathbb{R}^{d-1}$. The optimizer space is

$$\mathcal{O}_{\mathrm{BL},\Phi_S} = \big\{a \cdot \nabla\Phi_S + b : a \in \mathbb{R}^d, b \in \mathbb{R}\big\} = \mathrm{span}_{\mathbb{R}}\big\{1,\, \phi_S'(x),\, y_1, \dots, y_{d-1}\big\},$$

a $(d+1)$-dimensional subspace of $L^2(\rho_{\Phi_S})$.

Set $F_S(x,y) := f_S(x)$, so $\nabla F_S = (f_S'(x), 0_{\mathbb{R}^{d-1}})$ and $(D^2\Phi_S)^{-1}\nabla F_S = (f_S'(x)/\phi_S''(x), 0)$. Therefore

$$\big\langle (D^2\Phi_S)^{-1}\nabla F_S, \nabla F_S\big\rangle = \frac{(f_S'(x))^2}{\phi_S''(x)} \qquad \text{pointwise.}$$

Fubini and $\int d\gamma_{d-1} = 1$ give

$$\int_{\mathbb{R}^d} \big\langle (D^2\Phi_S)^{-1}\nabla F_S, \nabla F_S\big\rangle\, d\rho_{\Phi_S} = \int_{\mathbb{R}} \frac{(f_S')^2}{\phi_S''}\, d\rho_S = \mathcal{E}_{\phi_S}(f_S),$$

$$\int F_S\, d\rho_{\Phi_S} = \int f_S\, d\rho_S = 0, \qquad \mathrm{Var}_{\rho_{\Phi_S}}(F_S) = \int f_S^2\, d\rho_S = \mathrm{Var}_{\rho_S}(f_S),$$

hence $\delta_{\mathrm{BL},\Phi_S}(F_S) = \delta_{\mathrm{BL},\phi_S}(f_S)$.

Orthogonality of F_S to each generator of $\mathcal{O}_{\mathrm{BL},\Phi_S}$:

$$\langle F_S, 1\rangle_{L^2(\rho_{\Phi_S})} = \int f_S\, d\rho_S = 0,$$

$$\langle F_S, \phi_S'\rangle_{L^2(\rho_{\Phi_S})} = \Big(\int f_S\, \phi_S'\, d\rho_S\Big)\Big(\int d\gamma_{d-1}\Big) = 0 \cdot 1 = 0 \quad \text{by Lemma 3.1,}$$

$$\langle F_S, y_j\rangle_{L^2(\rho_{\Phi_S})} = \Big(\int f_S\, d\rho_S\Big)\Big(\int y_j\, d\gamma_{d-1}\Big) = 0 \cdot 0 = 0, \quad j = 1, \dots, d-1.$$

Hence $F_S \perp \mathcal{O}_{\mathrm{BL},\Phi_S}$ in $L^2(\rho_{\Phi_S})$, so

$$\mathrm{dist}_{L^2(\rho_{\Phi_S})}(F_S, \mathcal{O}_{\mathrm{BL},\Phi_S})^2 = \|F_S\|^2_{L^2(\rho_{\Phi_S})} = \|f_S\|^2_{L^2(\rho_S)},$$

and the divergent ratio of Theorem 3.3 is unchanged when one passes from (ϕ_S, f_S) to (Φ_S, F_S). This proves Theorem 1.1 in every dimension $d \geq 1$. □

Remark 3.4 (On the failure of the Poincaré route). The Brascamp–Lieb Dirichlet form is invariant under affine reparametrizations of the underlying space: for $T(x) = Ax + b$ with $A \in GL_d(\mathbb{R})$, set $\widetilde{\phi} := \phi \circ T^{-1}$ and $\widetilde{f} := f \circ T^{-1}$. Then $\rho_{\widetilde{\phi}} = T_{\#}\rho_\phi$, $D^2\widetilde{\phi}(y) = A^{-T}D^2\phi(T^{-1}y)A^{-1}$, $\nabla\widetilde{f}(y) = A^{-T}\nabla f(T^{-1}y)$, and a direct computation yields

$$\mathcal{E}_{\widetilde{\phi}}(\widetilde{f}) = \mathcal{E}_\phi(f), \qquad \mathrm{Var}_{\rho_{\widetilde{\phi}}}(\widetilde{f}) = \mathrm{Var}_{\rho_\phi}(f), \qquad \mathcal{O}_{\mathrm{BL},\widetilde{\phi}} = \mathcal{O}_{\mathrm{BL},\phi} \circ T^{-1}.$$

The pushforward L^2-distance to the optimizer space is also preserved: $\mathrm{dist}_{L^2(\rho_{\widetilde{\phi}})}(\widetilde{f}, \mathcal{O}_{\mathrm{BL},\widetilde{\phi}}) = \mathrm{dist}_{L^2(\rho_\phi)}(f, \mathcal{O}_{\mathrm{BL},\phi})$. Therefore the ratio $\mathrm{dist}^2_{L^2}/\delta_{\mathrm{BL}}$ is an affine invariant of (ϕ, f).

The Euclidean Poincaré constant $\lambda_1(\rho_\phi)^{-1}$, by contrast, *is not* affinely invariant: stretching a measure by an anisotropic diagonal matrix multiplies it by the squared anisotropy ratio. Any attempt to make the would-be L^2-stability constant blow up by exhibiting log-concave measures with unbounded ordinary Poincaré constant (such as long thin rectangles) is therefore doomed: the stretching that creates the large Poincaré constant is precisely the affine map under which the Brascamp–Lieb deficit, the L^2-distance to the optimizer space, and the optimizer space itself transform covariantly, leaving the relevant ratio unchanged.

The construction above evades this obstruction by creating a genuinely new near-eigenmode of the Brascamp–Lieb operator L_{ϕ_S}, in the orthogonal complement of $\mathcal{O}_{\mathrm{BL},\phi_S}$ rather than within an affine copy of it. Concretely, the even function f_S realizes a Rayleigh quotient $\mathcal{E}_{\phi_S}(f_S)/\operatorname{Var}_{\rho_S}(f_S) \to 1$ even though f_S stays orthogonal to the entire optimizer space $\{1, \phi_S'\}$; this reflects the spectral collapse described after Theorem 1.1 and is invariant under affine reparametrization.

References

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